Projections, Gram Matrices, and Statistical Information

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***Abstract*—This paper explores geometric connections among Gram matrices, orthogonal projections, and foundational statistical bounds, linking linear algebra with parameter estimation.**

I. Introduction

This note develops a unifying geometric perspective on several classical ideas in statistics and linear algebra. Starting from Gram matrices and orthogonal projections, we show how determinants, minors, and Schur complements encode the geometry of residuals and angles between vectors. These geometric quantities, in turn, determine the conditioning of linear systems, the strength of collinearity, and the precision with which individual parameters can be estimated.

The central thesis is simple. Principal minors of a Gram matrix measure the squared distance of one vector from the subspace spanned by the others; these residual distances are the quantities that determine angles, partial correlations, and the conditioning of the underlying system. In the statistical setting, the same geometry explains why nuisance parameters inflate the Cramér–Rao lower bound and why Gaussian graphical models encode conditional independence through zeros in the precision matrix.

The first part of the paper builds the core projection identity and derives the relation between principal minors, inverse Gram entries, and the angle between a vector and the subspace spanned by the others. We then connect these identities to the Cramér–Rao lower bound, partial correlation, and Gaussian graphical models, showing that the same geometry governs both algebraic stability and statistical efficiency.

II. Minors, angles, and the Cramér–Rao lower bound

A. Setting and notation

Let $v_1, \dots, v_n$ be vectors in a real inner product space $\mathcal{H}$, and let $G = [G_{ij}] \in \mathbb{R}^{n\times n}$ be their Gram matrix, defined by $G_{ij} = \langle v_i, v_j\rangle$. The space $\mathcal{H}$ may be finite-dimensional (such as Euclidean space) or infinite-dimensional (such as the L^2 space of random variables used later for Fisher information).

The following two definitions are used throughout.

Definition 1 (Gram matrix)**.** For vectors $v_1, \dots, v_r \in \mathcal{H}$, the Gram matrix is the $r \times r$ matrix

$$G^r = \mathrm{Gram}(v_1, \dots, v_r) = \big[\langle v_i, v_j\rangle\big]_{i,j=1}^{r}.$$

In particular, $G = \mathrm{Gram}(v_1, \dots, v_n)$.

Definition 2 (Bordered matrix)**.** If $A = \mathrm{Gram}(v_1, \dots, v_m)$, $b_i = \langle v_i, v_{m+1}\rangle$, and $\gamma = \|v_{m+1}\|^2$, the bordered Gram matrix obtained by appending v_{m+1} is

$$\mathrm{Bord}(A; b, \gamma) = \begin{pmatrix} A & b \\ b' & \gamma \end{pmatrix} = \mathrm{Gram}(v_1, \dots, v_m, v_{m+1}).$$

For this matrix, A is the leading principal block while b and γ capture cross-terms and self-energy of v.

For any $x \in \mathbb{R}^n$,

$$x'Gx = \sum_{i,j} x_i x_j \langle v_i, v_j\rangle = \Big\|\sum_i x_i v_i\Big\|^2 \geq 0 \tag{1}$$

Hence $G \succeq 0$ always, and $G \succ 0$ exactly when the v_k are linearly independent. We assume linear independence throughout, so G is invertible. Every principal submatrix of G is itself a Gram matrix of a subfamily and is therefore positive definite. Consequently, all principal minors are strictly positive, so the ratios below are well defined and nonzero.

Write M_{ij} for the first minor obtained by deleting row i and column j, and $C_{ij} = (-1)^{i+j} M_{ij}$ for the corresponding cofactor. The inverse entries satisfy $(G^{-1})_{ij} = C_{ji}/\det G = C_{ij}/\det G$, where the last equality follows because G, and therefore its cofactor matrix, is symmetric.

For $v \neq 0$ and a subspace $S \subseteq \mathcal{H}$, define the angle $\theta(v, S) \in [0, \pi/2]$ by

$$\cos\theta = \frac{\|P_S v\|}{\|v\|},$$

where P_S is the orthogonal projection onto S. Since $v = P_S v + (v - P_S v)$ is an orthogonal decomposition, the Pythagorean theorem gives $\|v\|^2 = \|P_S v\|^2 + \|v - P_S v\|^2$, and therefore

$$\sin\theta(v, S) = \frac{\mathrm{dist}(v, S)}{\|v\|}, \quad \mathrm{dist}(v, S) \coloneqq \|v - P_S v\|. \tag{2}$$

B. The projection lemma

The next lemma is the core computational step for this section. It links orthogonal projection, the Schur complement, and determinant factorization, with the Schur complement representing residual geometry.

Lemma 1 (Projection, Schur complement, and bordered determinant)**.** *Let $v_1, \dots, v_m$ be linearly independent with Gram matrix $A \in \mathbb{R}^{m\times m}$. Let $u \in \mathcal{H}$, and define the cross-terms $b_i \coloneqq \langle v_i, u\rangle$, the energy $\gamma \coloneqq \|u\|^2$, and the subspace $S \coloneqq \mathrm{span}\{v_1, \dots, v_m\}$. Then*

(i) *The projection of u onto S is $P_S u = \sum_i c_i v_i$ where the coefficient vector is $c = A^{-1} b$;*

(ii) *The squared norm of the projection is* $\|P_S u\|^2 = b'A^{-1}b$*, and the squared distance to the subspace is*

$$\operatorname{dist}^2(u,S) = \gamma - b'A^{-1}b := s, \tag{3}$$

which is exactly the Schur complement of A *in the bordered Gram matrix* $\bar{G} := \begin{pmatrix} A & b \\ b' & \gamma \end{pmatrix}$ *of* $(v_1,\ldots,v_m,u)$*;*

(iii) *The determinant of the bordered matrix scales by this distance:* $\det \bar{G} = s \det A$.

Proof. (i) The projection is uniquely characterized by finding a vector $\hat{u} \in S$ such that the error $u - \hat{u}$ is orthogonal to S. Orthogonality to S is equivalent to orthogonality to each basis vector v_i. Writing $\hat{u} = \sum_i c_i v_i$, the normal equations become:

$$\begin{aligned} 0 &= \langle u - \hat{u}, v_j \rangle \\ &= b_j - \sum_i c_i \langle v_i, v_j \rangle \\ &= b_j - (Ac)_j, \quad j = 1,\ldots,m, \end{aligned} \tag{4}$$

which simplifies to the matrix equation $Ac = b$. Since $A \succ 0$ is invertible, $c = A^{-1}b$ exists and is uniquely determined.

(ii) Using the relation $Ac = b$ twice, we find the squared norm of the projection:

$$\|\hat{u}\|^2 = c'Ac = c'b = b'A^{-1}b.$$

Since the error vector $u - \hat{u}$ is orthogonal to the projection $\hat{u}$, we apply the Pythagorean theorem to find the squared distance to the subspace:

$$\operatorname{dist}^2(u,S) = \|u-\hat{u}\|^2 = \langle u - \hat{u}, u \rangle = \gamma - c'b = \gamma - b'A^{-1}b.$$

(Expanding algebraically as $\gamma - 2c'b + c'Ac = \gamma - 2b'A^{-1}b + b'A^{-1}b$ gives the same value.) Note that $s > 0$ if and only if $u \notin S$, which matches the linear-independence requirement for the full extended family.

(iii) We can verify the block LDL' factorization (the matrix equivalent of completing the square) by multiplying it out:

$$\bar{G} = \begin{pmatrix} I & 0 \\ b'A^{-1} & 1 \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & s \end{pmatrix} \begin{pmatrix} I & A^{-1}b \\ 0 & 1 \end{pmatrix} \tag{5}$$

The right two factors produce $\left(\begin{smallmatrix} A & b \\ 0 & s \end{smallmatrix}\right)$. Left-multiplying this by the first unit lower-triangular matrix reproduces the $(2,1)$ entry $b'A^{-1}A = b'$ and the $(2,2)$ entry $b'A^{-1}b + s = \gamma$. Since the outer factors are unit triangular, their determinants are exactly 1. By the multiplicativity of determinants, we obtain $\det \bar{G} = s \det A$. $\square$

C. The diagonal minor ratio

Proposition 1 (Diagonal minor ratio)**.** *Let* θ_i *denote the angle between* v_i *and the subspace* $S_{-i} := \operatorname{span}\{v_k\}_{k \neq i}$ *spanned by all other vectors. Then*

$$\frac{\det G}{M_{ii}} = \operatorname{dist}^2(v_i, S_{-i}) = \|v_i\|^2 \sin^2\theta_i = G_{ii}\sin^2\theta_i, \tag{6}$$

and consequently, the diagonal entries of the inverse Gram matrix isolate this angular dependency:

$$(G^{-1})_{ii} = \frac{M_{ii}}{\det G} = \frac{1}{G_{ii}\sin^2\theta_i}, \quad \sin^2\theta_i = \frac{1}{G_{ii}\,(G^{-1})_{ii}} \tag{7}$$

Proof. Let P be the permutation matrix of the transposition (i,n). Then PGP' is the Gram matrix of the family with v_i and v_n interchanged. Because $\det P = \pm 1$, we have $\det(PGP') = (\det P)^2 \det G = \det G$. Deleting the last row and column of PGP' leaves the Gram matrix of $\{v_k\}_{k\neq i}$, whose determinant is M_{ii}; this is unaffected by the internal ordering of the remaining vectors. Hence, we may assume without loss of generality that $i = n$.

Apply Lemma 1 with $A = G^{n-1} = \operatorname{Gram}(v_1,\ldots,v_{n-1})$, $b_i = \langle v_i, v_n \rangle$ for $i = 1,\ldots,n-1$, $\gamma = \|v_n\|^2 = G_{nn}$, $S = S_{-n}$, $\bar{G} = G$, and $M_{nn} = \det G^{n-1}$.

Part (iii) gives $\det G = M_{nn}s$, and Part (ii) identifies $s = \operatorname{dist}^2(v_n, S_{-n})$, which by (2) equals $G_{nn}\sin^2\theta_n$. This establishes the first equation.

For the second equation, recall from $G^{-1} = \frac{1}{\det G}\operatorname{adj}(G)$ that $(G^{-1})_{nn} = C_{nn}/\det G$. Since $C_{nn} = (-1)^{2n}M_{nn} = M_{nn}$, we directly get $(G^{-1})_{nn} = M_{nn}/\det G = 1/s$. $\square$

The next two remarks provide consistency checks and clarify the statistical meaning of the result.

Remark 1 (Cauchy–Schwarz and the range of the ratio)**.** Applying the Cauchy–Schwarz inequality in $\mathbb{R}^n$ to the dual pair $G^{1/2}e_i$ and $G^{-1/2}e_i$,

$$\begin{aligned} 1 = e_i'e_i &= \langle G^{1/2}e_i, G^{-1/2}e_i \rangle \\ &\le \|G^{1/2}e_i\|\|G^{-1/2}e_i\| = \sqrt{G_{ii}(G^{-1})_{ii}}, \end{aligned} \tag{8}$$

so $G_{ii}(G^{-1})_{ii} \ge 1$, confirming that our rearranged equation indeed returns a valid geometric sine $\sin^2\theta_i \le 1$. Equality forces $G^{1/2}e_i \parallel G^{-1/2}e_i$, meaning $Ge_i = \lambda e_i$. In other words, the i-th column of G vanishes off the diagonal, implying $v_i \perp S_{-i}$ and $\theta_i = \pi/2$ (perfect orthogonality).

Remark 2 (Multiple correlation and VIF)**.** Let $g_k := \langle v_k, v_i \rangle$ for $k \neq i$, and let G^{-i} be the Gram matrix of $\{v_k\}_{k\neq i}$. Lemma 1(ii) gives $\|P_{S_{-i}}v_i\|^2 = g'(G^{-i})^{-1}g$. The squared multiple correlation—which measures how much variance of v_i is explained by the remaining vectors—is:

$$R_i^2 := \frac{\|P_{S_{-i}}v_i\|^2}{\|v_i\|^2} = \frac{g'(G^{-i})^{-1}g}{G_{ii}} = \cos^2\theta_i, \tag{9}$$

hence

$$\sin^2\theta_i = 1 - R_i^2$$

In regression diagnostics, the factor $1/(1-R_i^2)$ is recognized as the Variance Inflation Factor (VIF). Geometrically, as the vector v_i becomes highly collinear with the other predictors ($R_i^2 \to 1$), the angle θ_i approaches zero. The VIF measures exactly how much the variance of an estimated regression coefficient increases due to this collinearity compared to an orthogonal baseline.

Corollary 1 (Sequential factorization and Hadamard's inequality)**.** *Let* $V^{k-1} := \operatorname{span}\{v_1,\ldots,v_{k-1}\}$ *(with* $V^0 = \{0\}$*) and* $\phi_k := \theta(v_k, V^{k-1})$*. Then the total volume of the parallelotope formed by the vectors is bounded by the product of their lengths:*

$$\det G = \prod_{k=1}^n \operatorname{dist}^2(v_k, V^{k-1}) = \prod_{k=1}^n G_{kk}\sin^2\phi_k \le \prod_{k=1}^n G_{kk} \tag{10}$$

Proof. We use induction on n with Lemma 1(iii). The Gram matrix of $v_1, \dots, v_k$ is the bordering of the Gram matrix of $v_1, \dots, v_{k-1}$ by v_k. Thus, its determinant is the previous determinant multiplied by the squared orthogonal distance $\operatorname{dist}^2(v_k, V^{k-1})$. The base case $n = 1$ is trivially $\det G = \|v_1\|^2$. The inequality follows from $\sin^2 \phi_k \le 1$, with equality holding throughout if and only if the family is mutually orthogonal. $\square$

There is a fundamental asymmetry between (10) and (6): the angles ϕ_k are *sequential* (each vector is evaluated only against its predecessors and therefore depends strongly on ordering), whereas θ_i is defined against *all* other vectors. Only θ_i is permutation invariant, and only it answers how independently coordinate i is determined.

D. Off-diagonal entries: the dual basis and partial correlation

Remark 3 (Dual basis and residuals)**.** Let $r_i := v_i - P_{S_{-i}} v_i$ be the residual from Proposition 1, and define the dual family $w_i := \sum_k (G^{-1})_{ik} v_k$. The inner products of this dual basis with the primal basis yield the Kronecker delta:

$$
\begin{aligned}
\langle w_i, v_j \rangle &= \sum_k (G^{-1})_{ik} \langle v_k, v_j \rangle = \sum_k (G^{-1})_{ik} G_{kj} \\
&= (G^{-1} G)_{ij} = \delta_{ij}, && (11)\\
\langle w_i, w_j \rangle &= \sum_{k,\ell} (G^{-1})_{ik} G_{k\ell} (G^{-1})_{\ell j} \\
&= (G^{-1} G G^{-1})_{ij} = (G^{-1})_{ij}, && (12)
\end{aligned}
$$

showing that G^{-1} is itself the Gram matrix of the dual basis. Moreover, the dual basis vectors are proportional to the residuals:

$$w_i = \frac{r_i}{\|r_i\|^2} \tag{13}$$

To see this, note that $\langle w_i, v_j \rangle = \delta_{ij}$ means $w_i \perp v_j$ for every $j \neq i$, hence $w_i \perp S_{-i}$. Also $w_i \in V := \operatorname{span}\{v_1, \dots, v_n\}$. Since $\dim V - \dim S_{-i} = 1$, the orthogonal complement of S_{-i} inside V is a single line, which contains both w_i and the non-zero residual $r_i \neq 0$. Thus $w_i = c r_i$. (12) yields $c = 1/\|r_i\|^2$ is the normolization factor.

Taking $j = i$ in (12) recovers $(G^{-1})_{ii} = 1/\|r_i\|^2$, in agreement with Proposition 1. Taking $j \neq i$ and normalizing yields the geometric cosine between residuals:

$$\frac{(G^{-1})_{ij}}{\sqrt{(G^{-1})_{ii} (G^{-1})_{jj}}} = \frac{\langle r_i, r_j \rangle}{\|r_i\| \|r_j\|} = \cos\angle(r_i, r_j) \tag{14}$$

Proposition 2 (Off-diagonal ratio and partial correlation)**.** *Fix $i \neq j$, let $S_{-i,-j} := \operatorname{span}\{v_k\}_{k \neq i,j}$ (the "control" subspace), and let $\tilde r_i := v_i - P_{S_{-i,-j}} v_i$ and $\tilde r_j := v_j - P_{S_{-i,-j}} v_j$ be the residuals after projecting out only the other $n-2$ vectors. Then the partial correlation—the correlation between v_i and v_j after removing the linear effects of all other variables—is:*

$$
\begin{aligned}
\rho_{ij|\mathrm{rest}} &:= \cos\angle(\tilde r_i, \tilde r_j) \\
&= \frac{-(G^{-1})_{ij}}{\sqrt{(G^{-1})_{ii} (G^{-1})_{jj}}} \qquad (15)\\
&= \frac{-(-1)^{i+j} M_{ij}}{\sqrt{M_{ii} M_{jj}}}
\end{aligned}
$$

Proof. It suffices to take $i = n-1$ and $j = n$ by reordering the system. Partition the matrix into blocks accordingly:

$$
\begin{gathered}
G = \begin{pmatrix} A & B \\ B' & D \end{pmatrix}, \quad A \in \mathbb{R}^{(n-2)\times(n-2)}, \\
B_{k,n-1} = \langle v_k, v_{n-1} \rangle, \quad B_{k,n} = \langle v_k, v_n \rangle, \\
D_{n-1,n-1} = \langle v_{n-1}, v_{n-1} \rangle,\ D_{n-1,n} = \langle v_{n-1}, v_n \rangle,\ D_{n,n} = \langle v_n, v_n \rangle
\end{gathered}
$$

where k ranges over the $n-2$ remaining indices. Block inversion (the 2×2 block version of the LDL$'$ factorization) dictates that the trailing block of G^{-1} is the inverse of the Schur complement:

$$[(G)^{-1}]_{\{n-1,n\}} = H^{-1}, \qquad H := D - B' A^{-1} B \tag{16}$$

Here, H is precisely the Gram matrix of $(\tilde r_{n-1}, \tilde r_n)$. By Lemma 1(i), $P_{S_{-i,-j}} v_{n-1} = \sum_k c_k^{(n-1)} v_k$ with $c^{(n-1)} = A^{-1} B_{\cdot,n-1}$. Because the error $\tilde r_{n-1}$ is orthogonal to the control subspace $S_{-i,-j}$ while the projection $v_n - \tilde r_n \in S_{-i,-j}$, we get:

$$
\begin{aligned}
\langle \tilde r_{n-1}, \tilde r_n \rangle &= \langle \tilde r_{n-1}, v_n \rangle \\
&= D_{n-1,n} - (c^{(n-1)})' B_{\cdot,n} \\
&= D_{n-1,n} - B'_{\cdot,n-1} A^{-1} B_{\cdot,n} \\
&= H_{n-1,n}
\end{aligned}
$$

For a 2×2 symmetric matrix H, the adjugate matrix is $\left(\begin{smallmatrix} H_{22} & -H_{12} \\ -H_{12} & H_{11} \end{smallmatrix}\right)$. Thus, $\det H$ naturally cancels out in the normalized ratio:

$$\frac{-(H^{-1})_{12}}{\sqrt{(H^{-1})_{11} (H^{-1})_{22}}} = \frac{H_{12}/\det H}{\sqrt{H_{11} H_{22}}/\det H} = \frac{\langle \tilde r_{n-1}, \tilde r_n \rangle}{\|\tilde r_{n-1}\| \|\tilde r_n\|}$$

Substituting (16) and rewriting $(G^{-1})_{ij} = C_{ij}/\det G$ in cofactor form yields (15). $\square$

The minus sign in (15) is not an arbitrary statistical convention: it comes directly from the off-diagonal sign of the 2×2 adjugate. Comparing (14) with (15) reveals a geometric sign flip:

$$\cos\angle(r_i, r_j) = -\cos\angle(\tilde r_i, \tilde r_j). \tag{17}$$

This highlights that the residuals in Proposition 1 are *not* the same as those in Proposition 2. Additionally, projecting v_j out of v_i (and vice versa) reverses the sign of the cosine, and the leading minus sign in the partial-correlation formula compensates for this reversal.

Example 4 ($n = 2$)**.** For a basic bivariate system $G = \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$, we compute the inverse $G^{-1} = (1-\rho^2)^{-1} \left(\begin{smallmatrix} 1 & -\rho \\ -\rho & 1 \end{smallmatrix}\right)$. Proposition 2 returns $\rho_{12} = \rho$ (as there are no other variables S_{ij} to partial out).

Meanwhile, the individual residuals from projecting onto each other are $r_1 = v_1 - \rho v_2$ and $r_2 = v_2 - \rho v_1$. These satisfy $\langle r_1, r_2 \rangle = -\rho(1-\rho^2)$ and $\|r_k\|^2 = 1 - \rho^2$, giving $\cos\angle(r_1, r_2) = -\rho$, which illustrates the sign flip in (17). We also observe $\sin^2 \theta_1 = 1 - \rho^2$ and $(G^{-1})_{11} = 1/(1-\rho^2)$, matching the VIF formulation.

E. The Cramér–Rao lower bound

We now translate this geometry into statistical estimation and quantify how correlated nuisance parameters limit attainable accuracy.

For a parametric model with density $p(x;\vartheta)$, the likelihood function is

$$L(\vartheta;x) \coloneqq p(x;\vartheta), \tag{18}$$

and the corresponding log-likelihood is

$$\ell(\vartheta;x) \coloneqq \log L(\vartheta;x) = \log p(x;\vartheta). \tag{19}$$

The score function with respect to the i-th parameter is then

$$\dot{\ell}_i(x;\vartheta) \coloneqq \partial_i \log p(x;\vartheta), \tag{20}$$

which measures how sensitive the log-likelihood is to a small change in ϑ_i. The collection of score functions forms a family of random variables in $L^2(p)$, and their covariance structure is precisely the Fisher information matrix.

Definition 3 (Cramér–Rao lower bound)**.** Let $\vartheta \in \mathbb{R}^n$ parameterize a statistical model with Fisher information matrix $\mathbf{F}(\vartheta)$. For any unbiased estimator $\hat{\vartheta}$ of ϑ, the covariance matrix satisfies

$$\operatorname{cov}(\hat{\vartheta}) \succeq \mathbf{F}(\vartheta)^{-1}$$

In particular, the variance of each component obeys

$$\operatorname{var}(\hat{\vartheta}_i) \geq (\mathbf{F}^{-1})_{ii}$$

and the right-hand side is called the Cramér–Rao lower bound (CRLB) for ϑ_i.

Corollary 2 (Nuisance penalty in the Cramér–Rao lower bound)**.** *Let $\mathbf{F}$ be the Fisher information matrix for a parameter vector $\vartheta \in \mathbb{R}^n$. By definition, $\mathbf{F}_{ij} = \mathrm{E}\big[\partial_i \log p \partial_j \log p\big]$ is the Gram matrix of the score functions $\dot{\ell}_i \coloneqq \partial_i \log p$ in the Hilbert space $L^2(p)$ under the inner product $\langle a,b\rangle \coloneqq \mathrm{E}[ab]$. Then the minimum variance of an unbiased estimator is bounded by:*

$$\begin{aligned}\mathrm{CRLB}(\vartheta_i) &= (\mathbf{F}^{-1})_{ii} = \frac{M_{ii}}{\det \mathbf{F}} \\ &= \underbrace{\frac{1}{\mathbf{F}_{ii}}}_{\substack{\text{single-parameter}\\ \text{bound}}} \cdot \underbrace{\frac{1}{\sin^2\theta_i}}_{\substack{\text{nuisance}\\ \text{penalty}}} = \frac{1}{\mathbf{F}_{ii}(1-R_i^2)}\end{aligned} \tag{21}$$

Here, θ_i is the angle in $L^2(p)$ between the i-th score and the span of the nuisance scores. The penalty equals unity if and only if the i-th score is orthogonal to all others (parameter orthogonality). It diverges as $\csc^2\theta_i$ in the ill-conditioned regime where $\det \mathbf{F} \to 0$ while M_{ii} is bounded away from zero.

Proof. Under usual regularity conditions (e.g., the density $p_\vartheta > 0$, ability to swap differentiation and integration), the score functions lie in $L^2(p)$ with a mean of zero, $\mathrm{E}[\dot{\ell}_i] = 0$. Consequently, $\mathbf{F}$ serves simultaneously as their covariance matrix and their geometric Gram matrix. The result follows directly by substituting $v_i = \dot{\ell}_i$ into Proposition 1 and utilizing the multiple correlation equivalence. □

Remark 5 (Efficient score)**.** Let $\dot{\ell}_i^* \coloneqq \dot{\ell}_i - P_{S_i}\dot{\ell}_i$ be the residual of the i-th score after orthogonally projecting out the nuisance scores. This is known in semiparametric theory as the *efficient score* for the parameter component ϑ_i. By Proposition 1:

$$\mathrm{CRLB}(\vartheta_i) = \frac{1}{\|\dot{\ell}_i^*\|^2} = \frac{1}{\mathrm{E}\big[(\dot{\ell}_i^*)^2\big]} \tag{22}$$

This highlights a key statistical point: the achievable accuracy of estimating ϑ_i is determined not by the total sensitivity of the likelihood to ϑ_i, but *only* by the part of that sensitivity that nuisance parameters cannot mimic. It also shows that the penalty is a property of the *subspace* S_i, not of a particular parametrization of the nuisance block. Any reparametrization that keeps ϑ_i fixed while re-coordinatizing nuisance variables leaves S_i—and therefore ϑ_i and the optimal bound—unchanged.

Example 6 (Linear Gaussian model / collinearity)**.** Let $y = A\vartheta + \varepsilon$ with $\varepsilon - \mathcal{N}(0,\sigma^2 I)$ and $A = [a_1,\ldots,a_n]$ having full column rank. Then the log-likelihood is $\log p = -\|y - A\vartheta\|^2/(2\sigma^2) + \text{const}$. The score functions take the form $\dot{\ell}_i = a_i'(y - A\vartheta)/\sigma^2$, leading to:

$$\mathbf{F}_{ij} = \frac{a_i' \; \mathrm{E}\big[(y-A\vartheta)(y-A\vartheta)'\big]\, a_j}{\sigma^4} = \frac{a_i' a_j}{\sigma^2}$$

Thus, the Fisher information matrix is the regressor Gram matrix $A'A$ scaled by the noise variance. In this context, the parameter component ϑ_i is associated with the geometric angle in $\mathbb{R}^m$ between the column vector a_i and the span of the remaining columns. Consequently:

$$\mathrm{CRLB}(\vartheta_i) = \frac{\sigma^2}{\|a_i\|^2 \sin^2\theta_i} = \frac{\sigma^2}{\|a_i\|^2} \cdot \frac{1}{1-R_i^2} \tag{23}$$

Collinearity appears as a small angle. The regressor energy $\|a_i\|^2$ (representing the signal strength of that variable) may be large, but if the vector lies too close to the hyperplane spanned by the other variables, the parameter estimate remains unreliable.

Note on Regularization: This geometric collapse is exactly why techniques like Ridge regression (Tikhonov regularization) add a penalty term λI to the Gram matrix. Geometrically, this artificially inflates the lengths and angles, bounding the variance at the cost of introducing bias. The exact same geometry governs closely spaced components in superresolution problems (like separating two closely spaced point sources in optics), where $\sin\theta_i \to 0$ as separation shrinks, causing the theoretical error bounds to blow up even at fixed signal-to-noise ratios.

Remark 7 (Diagnostics and computational stability)**.** The Cramér–Rao formulation clearly shows that the determinant $\det \mathbf{F}$ alone—which is maximized in the D-optimality design criterion—cannot successfully pinpoint *which* parameter is poorly determined. By Corollary 1, the determinant aggregates the sequential angles into a single overall volume metric. A small parallelotope volume communicates that the system is ill-conditioned, but says nothing about which coordinate is failing.

The true diagnostic quantity is the ratio $M_{ii}/\det \mathbf{F}$, not either factor in isolation. Numerically, however, calculating explicit determinants and minors is disastrous due to floating-point underflow/overflow and $O(n!)$ naive complexity.

Instead, computationally stable diagnostics compute the Cholesky factorization $\mathbf{F} = LL'$ in $O(n^3)$ operations. One can then obtain the inverse diagonal entries $(\mathbf{F}^{-1})_{ii} = \|L^{-1}e_i\|^2$ via standard forward/backward triangular solves, and finally read off the geometric angles using $\sin^2\theta_i = 1/(\mathbf{F}_{ii}(\mathbf{F}^{-1})_{ii})$.

III. Application to Gaussian random vectors

The projection-and-Gram framework has a natural application to multivariate normal models. With an appropriate inner product on random variables, algebraic identities for minors and inverses become statistical statements about conditional dependence and optimal prediction.

A. The geometry of conditional expectation

Let $X = \left[X_1, \ldots, X_n\right]'$ be a random vector drawn from a multivariate normal distribution $\mathcal{N}(\mu, \Sigma)$, where we assume the covariance matrix $\Sigma \succ 0$ is strictly positive definite. Without loss of generality, we assume $\mu = 0$, as shifting the mean does not affect variances or covariances.

Consider the Hilbert space $L^2(P)$ of zero-mean, finite-variance random variables, equipped with the covariance inner product:

$$\langle U, V\rangle := \mathrm{E}[UV] = \mathrm{cov}(U, V) \tag{24}$$

Under this geometry, the covariance matrix Σ is exactly the Gram matrix G of the variables $X_1, \ldots, X_n$. The squared norm of a variable is its variance, $\|X_i\|^2 = \mathrm{var}(X_i) = \Sigma_{ii}$, and the cosine of the angle between two variables is exactly their Pearson correlation coefficient:

$$\cos\angle(X_i, X_j) = \frac{\langle X_i, X_j\rangle}{\|X_i\|\|X_j\|} = \frac{\mathrm{cov}(X_i, X_j)}{\sqrt{\mathrm{var}(X_i)\,\mathrm{var}(X_j)}} = \rho_{ij}. \tag{25}$$

For Gaussian random vectors, a remarkable property holds: orthogonal projection of a variable onto a subspace generated by other variables is equivalent to its *conditional expectation*.

If $S_{-i} := \mathrm{span}\{X_k\}_{k\neq i}$ is the subspace spanned by all other variables (denoted collectively as X_{-i}), the projection is:

$$P_{S_{-i}} X_i = \mathrm{E}[X_i \mid X_{-i}] \tag{26}$$

Because the projection is linear, the conditional expectation of a Gaussian is always a linear combination of the conditioning variables.

B. The precision matrix and conditional variance

Let $\Theta := \Sigma^{-1}$ be the inverse covariance matrix, also called the *precision matrix*. Applying Proposition 1 in this setting gives a direct statistical interpretation of its diagonal entries.

Corollary 3 (Diagonal of the precision matrix)**.** *The squared distance from X_i to the subspace S_{-i} represents the variance of the residual error, which is exactly the conditional variance of X_i given all other variables:*

$$\mathrm{dist}^2(X_i, S_{-i}) = \mathrm{E}\big[(X_i - \mathrm{E}[X_i \mid X_{-i}])^2\big] = \mathrm{var}(X_i \mid X_{-i}) \tag{27}$$

Following Proposition 1, the diagonal entries of the precision matrix isolate this conditional variance:

$$\Theta_{ii} = (\Sigma^{-1})_{ii} = \frac{1}{\mathrm{var}(X_i \mid X_{-i})} \tag{28}$$

Using the angular formulation from Proposition 1, we can also express this as:

$$\mathrm{var}(X_i \mid X_{-i}) = \mathrm{var}(X_i)\sin^2\theta_i = \mathrm{var}(X_i)(1 - R_i^2) \tag{29}$$

where R_i^2 is the proportion of variance of X_i explained by all other variables. If X_i is highly correlated with the rest of the network ($\sin\theta_i \to 0$), the conditional variance shrinks toward zero, and the precision Θ_{ii} blows up to infinity.

C. Partial correlation and Gaussian graphical models

We now apply Proposition 2 to interpret the off-diagonal entries of the precision matrix. Let $\tilde{r}_i$ and $\tilde{r}_j$ be the residuals of X_i and X_j after projecting out the control subspace $S_{-i,-j} := \mathrm{span}\{X_k\}_{k\neq i,j}$. In probability terms, these are the prediction errors:

$$\tilde{r}_i = X_i - \mathrm{E}[X_i \mid X_{-i,-j}], \quad \tilde{r}_j = X_j - \mathrm{E}[X_j \mid X_{-i,-j}] \tag{30}$$

The correlation between these residuals is the *partial correlation*, denoted $\rho_{ij\cdot\mathrm{rest}}$. It measures the linear dependence between X_i and X_j that cannot be explained by the other variables.

Corollary 4 (Off-diagonal precision and partial correlation)**.** *The partial correlation between X_i and X_j given all other variables is given by the scaled off-diagonal entries of the precision matrix:*

$$\rho_{ij\setminus\mathrm{rest}} = \frac{-\Theta_{ij}}{\sqrt{\Theta_{ii}\Theta_{jj}}} \tag{31}$$

This equation underpins Gaussian graphical models (GGMs): for jointly Gaussian variables, zero correlation implies statistical independence. Therefore, if the partial correlation is zero, the residuals are independent, meaning X_i and X_j are *conditionally independent* given the rest of the network:

$$\Theta_{ij} = 0 \iff X_i \perp\!\!\!\perp X_j \mid X_{-i,-j} \tag{32}$$

Geometrically, $\Theta_{ij} = 0$ means that after projecting out the shared environment $S_{-i,-j}$, the remaining variations of X_i and X_j are orthogonal.

D. Information geometry and entropy

Finally, Corollary 1 can be interpreted in information-theoretic terms. The differential entropy of an n-dimensional Gaussian vector is:

$$h(X) = \frac{1}{2}\log\det(2\pi e\Sigma) = \frac{n}{2}\log(2\pi e) + \frac{1}{2}\log\det\Sigma \tag{33}$$

Because $\det \Sigma$ factors sequentially into the squared distances (conditional variances), we have:

$$\det \Sigma = \prod_{k=1}^{n} \mathrm{var}(X_k \mid X_1, \ldots, X_{k-1}) \tag{34}$$

Taking the logarithm transforms this geometric volume factorization directly into the chain rule of conditional entropy:

$$h(X_1, \ldots, X_n) = \sum_{k=1}^{n} h(X_k \mid X_1, \ldots, X_{k-1}) \tag{35}$$

Hadamard's inequality ($\det \Sigma \leq \prod \Sigma_{kk}$) geometrically states that the volume of a parallelotope is maximized when its sides are orthogonal. Information-theoretically, this is the well-known theorem that for a given set of marginal variances, the joint entropy is maximized when the variables are completely independent.